

Testing LLMs on JOKER

Task 3: Onomastic Wordplay Translation

Task 4: Humour Generation

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What did we test?

Task 3: translation

English name-based wordplay → French name.

Preserve the pun or cultural effect using a short description of the character.

Five prompted LLMs; no task-specific fine-tuning.

Task 4: generation

Structured pun context → a short joke.

Generate puns in English, Spanish and French.

Generate French *contrepèteries*.

Models and experimental setup

- ▶ **Gemini:** gemini-3-flash-preview
- ▶ **Open models:** Qwen3-4B-Instruct-2507, Mistral-7B-Instruct-v0.3, Gemma-2-9b-it, Meta-Llama-3.1-8B-Instruct
- ▶ One run per model for translation
- ▶ 5 runs per language for pun generation
- ▶ 5 runs for contrepetrie generation in French
- ▶ Same direct prompting approach across systems, with task-specific instructions

Task 3: 5 runs.

Task 4: 20 runs (15 puns + 5 French contrepèteries).

Task 3: data format

```
{  
  "id": "en_2",  
  "en": "Obelix",  
  "description": "Obélix is one of the main characters in the  
    → Asterix comics. He is a strong, large, and good-natured  
    → Gaul who, unlike other villagers, has permanent superhuman  
    → strength due to accidentally drinking a magic potion as a  
    → child. Obélix is best friends with Asterix, and together  
    → they embark on numerous adventures. Known for his love of  
    → wild boar, Obélix's loyalty, humor, and appetite make him  
    → a beloved character in the series",  
  "fr": "Obélix"  
}
```

Task 3: prompt

f"Translate the following English named entity into French.
This is a localization task, not a literal translation. Use the
→ description to infer the meaning, symbolism, personality, and
→ any hidden wordplay in the original name.
If the name contains a pun, portmanteau, joke, double meaning, or
→ sound-based wordplay, create a French name that produces a
→ similar effect for a French-speaking audience, even if the
→ wording differs significantly from the original.
Output only the final French name.
Named entity: \"{ENTITY}\"
Description: \"{DESCRIPTION}\""

Task 3: Gemini leads on valid translations

System	Official score	Reference match	Valid alternative	Copy
Gemini	71.88	44.11%	45.37%	8.20%
Llama	17.25	7.08%	10.54%	24.83%
Identity baseline	13.10	13.66%	0%	100%
Qwen	4.47	3.54%	0.32%	4.35%
Mistral	2.24	2.42%	1.60%	6.31%
Gemma	1.92	1.35%	0.64%	0.15%

- ▶ Gemini produces the most valid alternative translations.
- ▶ All prompted open models underperform the copy baseline.

Task 4: Data format

```
{  
  "id": "fr_0001",  
  "lang": "fr",  
  "pun_word": "barbec-ours",  
  "sense_a": "barbecue",  
  "sense_b": "bear",  
  "keywords": ["BBQ", "cookout", "grill", "grizzly", "brown  
    ↪ bear", "cub", "..."],  
  "hint": "barbec-ours / barbecue + ours ~ <côtelette; grizzly;  
    ↪ cuit>",  
  "reference_joke": "Comment on cuit des côtelette de grizzly ? Au  
    ↪ barbec-ours.",  
  "reference_en": "What do you call grizzly's ribs ? A barbecue."  
}
```

Task 4: Pun generation prompt (EN, FR, ES)

f"You are an expert pun writer in {LANGUAGE}.

Given the context below, generate one original, witty pun in

↪ {LANGUAGE} inspired by the main topic, concepts, or vocabulary

↪ in the context.

Rules:

- Make the {LANGUAGE} pun natural and funny.
- Keep it to one sentence.
- Avoid generic jokes.
- Do not include quotation marks, introductions, explanations, or
↪ any text other than the pun itself.

Context: {CONTEXT}"

Task 4: Contrepèterie generation in French

- ▶ A *contrepèterie* is a common form of French wordplay in which swapping sounds, letters, or syllables between words reveals a hidden humorous meaning
- ▶ *Quel beau **plan** de **foire** ! — Quel beau **flan** de **poire** ! (What a great fair plan! — What a beautiful pear flan!)*

Task 4: Contrepèterie generation prompt

f"Based on the provided context, generate **exactly one authentic French contrepèterie**.

Return **only** the following three fields, in this exact format:

Contrepèterie: <original French sentence>

Sens 1: <literal, innocent meaning of the original sentence>

Sens 2: <hidden meaning revealed by the contrepèterie>

Requirements:

- * The contrepèterie must be a genuine and natural-sounding French
↪ expression.
- * The original sentence must be grammatically correct and idiomatic
↪ French.
- * The hidden meaning must result from a valid contrepèterie (phoneme,
↪ syllable, or letter transposition commonly accepted in French
↪ wordplay).

Task 4: Evaluation

- ▶ The JOKER Lab Arena compares pairs of outputs using three LLM judges
- ▶ Pairwise preferences yield separate Bradley–Terry leaderboards for each language

Task 4: Results (EN,ES)

	Rank	Run	BT	CI low	CI high	n_b
English	6	gemini-3_pun	1,125	1,084	1,168	326
	71	qwen_pun	995	953	1,039	299
	87	mistral_pun	903	869	937	343
	89	gemma_pun	895	851	929	360
	92	llama_pun	816	765	858	316
Spanish	4	gemini-3_pun	1,351	1,310	1,396	363
	19	gemma_pun	1,125	1,084	1,172	322
	38	qwen_pun	1,100	1,072	1,144	366
	93	llama_pun	1,019	984	1,063	360
	106	mistral_pun	987	945	1,025	332

Task 4: Results (FR)

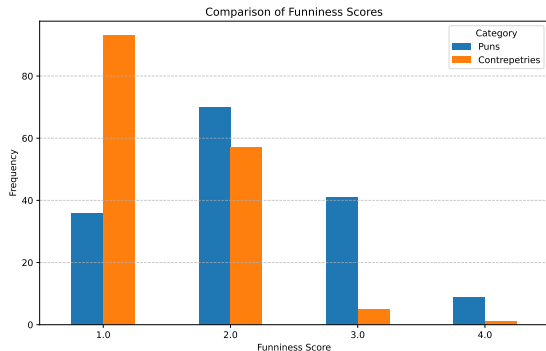
Rank	Run	BT	CI low	CI high	n_b
7	gemini-3-flash_pun	1,279	1,234	1,341	362
50	qwen-3_pun	1,130	1,092	1,165	410
55	llama-3_pun	1,036	989	1,077	342
59	gemma-3_pun	1,022	973	1,062	350
60	mistral-3_pun	1,021	982	1,052	325
117	mistral_contrepetrie	682	630	719	378
119	llama_contrepetrie	662	613	703	348
120	gemini-3-flash_contrepetrie	655	601	706	331
121	qwen-3_contrepetrie	644	577	694	356
123	gemma_contrepetrie	618	564	661	355

Manual analysis: 156 X 2 French outputs

Two native French-speaking Master's students in translation annotated Gemini outputs; each instance received one annotation.

Criterion (“yes”)	Puns	Contrepèteries
Wordplay	152/156	87/156
Pun	126/156	22/156
Valid contrepèterie	–	79/156
Related to the input pun word	119/156	39/156
Offensive	1/156	37/156

Funniness: the manual ratings



Generated jokes were generally rated as not funny.

About half of the contrepèteries received the minimum score (1 of 5).

Why does pun generation fail?

- ▶ **Unnecessary detail:** a pun may work linguistically but include words that do not strengthen the joke.
 - ▶ For *sense_a* “alaitchante” and *sense_b* “alléchante”, Gemini generated the joke “*La carrière d’une vache **soprano** est forcément alaitchante*”.
- ▶ **Generated text is not idiomatic:**
 - ▶ “*Difficile pour un moteur de tourner rond quand sa cylindrée est **restée bloquée au cube***”

Why does contrepètrie generation fail?

- ▶ **Attempting to resolve the intended sound or letter transpositions fails to produce a valid FR sentence**
 - ▶ “Goûtez-moi cette tomate” yields the nonsensical “**Toutez**-moi cette **gomate**”
- ▶ **Weak grounding:** 110 of 156 generated contrepèteries were judged unrelated to the input pun word.
- ▶ **Gemini occasionally reproduces well-known *contrepèteries*:**
 - ▶ “folle de la messe” appeared 9 times for different inputs (“Cedi, ” “il en va de suie”, “anagram”, “folie”, “cinq”, “formes”, “dong”, “obsolètement”, and “SI | Venn”)
 - ▶ “glisser dans la piscine” is generated 4 times for the unrelated “précipitation”, “Venn”, “transparent”, “White | Bluecorp”

Why do generated jokes fail?

- ▶ **Unnecessary detail:** a pun may work linguistically but include words that do not strengthen the joke.
- ▶ **Invalid transposition:** “Goûtez-moi cette tomate” yields the nonsensical “Toutez-moi cette gomite” under the intended swap.
- ▶ **Weak grounding:** 110 of 156 generated contrepèteries were judged unrelated to the input pun word.
- ▶ **Repeated stock examples:** “folle de la messe” appeared nine times for different inputs.

Conclusions

- ▶ Current LLMs still struggle with the linguistic constraints underlying wordplay
- ▶ Llama left 25% of source names untranslated, while Mistral, Qwen, and Gemma rarely did so but achieved limited success
- ▶ Gemini leads on name-based translation and ranks among the strongest pun generators in all three languages
- ▶ The best-performing systems remained below a 50% success rate for the cases requiring a genuine translation
- ▶ Authentic French contrepèteries remain difficult to generate and to evaluate automatically
- ▶ AI-generated puns are often overly verbose or include unnecessary details, which can reduce their humorous effect
- ▶ Gemini occasionally reproduces well-known *contrepèteries* instead of generating new ones

Thank you

Questions?

UBO at JOKER, CLEF 2026